

Manuel MADEIRA

PERSONAL DATA

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ABOUT ME

I am a 4th-year PhD student at EPFL, supervised by [Pascal Frossard](#) and [Dorina Thanou](#). My research focuses on the interplay between generative models, in particular diffusion, and the structure in data, exploring both how these models exploit inherent data structure and how to explicitly encode structure into them. Using graph generation as a testbed, I have studied how to incorporate domain knowledge into discrete diffusion and how to align discrete flow matching with graph-specific inductive biases. I apply these approaches to scientific domains, including molecular generation and digital pathology. I am also interested in extending them to broader domains, e.g., language.

EDUCATION

2022 - PRESENT	PhD. in MACHINE LEARNING (Expected Graduation: December 2026) École Polytechnique Fédérale de Lausanne , Switzerland
2018 - 2019 (FALL)	Exchange student in COMPUTER SCIENCE Tsinghua University , China
2018 - 2021	MSc. in BIOMEDICAL ENGINEERING Instituto Superior Técnico , Portugal GPA: 19 / 20 (1st in class), THESIS: 20/20
2015 - 2018	BSc. in BIOMEDICAL ENGINEERING Instituto Superior Técnico , Portugal GPA: 19 / 20 (1st in class and 1st ever to attain such grade)

RESEARCH EXPERIENCE

JUL 2025 - DEC 2025	Applied Scientist Intern at AMAZON Developed a graph-based deep learning framework achieving $\sim 1000\times$ efficiency improvements over existing tools for supply chain long-term planning.
SEP 2022 - PRESENT	Doctoral Assistant at EPFL Studied the incorporation of structural constraints into graph discrete diffusion models and developed the first discrete flow matching model for graphs. Employed these methods in real-world applications, including molecular generation and digital pathology.
SEP 2021 - AUG 2022	Machine Learning Researcher at INDUCTIVA RESEARCH LABS (Startup) Conducted research on deep learning based approaches to solve partial differential equations. Deployed physics-informed neural networks to model heat diffusion and coastal dynamics and analysed their generalization to arbitrary domains.
MAR 2021 - JUN 2021	Research Internship at INSTITUTE FOR SYSTEMS AND ROBOTICS SUPERVISORS: Renato Negrinho , João Xavier , and Pedro Aguiar Researched on L -smoothness exploitation for first-order stochastic optimization methods with theoretical developments on algorithmic analysis.
SEP 2019 - FEB 2020	Research Internship at IST-ID Transcribed interviews on health technologies assessment in Portugal for the MEDI-VALUE PROJECT .

AWARDS AND HONORS

2025	Top Reviewer , by NeurIPS 2025
2025	Qualcomm Innovation Fellowship Finalist , by Qualcomm
2022-23	EDIC PhD Fellowship , by EPFL
2016-18	1st ranked student in Biomedical Engineering BSc , by Instituto Superior Técnico
2015-20	Diploma of Academic Excellence (Top 10%) , by Instituto Superior Técnico
2009-15	Best student in high school , by Crédito Agrícola

PUBLICATIONS

[1] A. Carballo-Castro, MM, Y. Qin, D. Thanou, and P. Frossard. **Generating Directed Graphs with Dual Attention and Asymmetric Encoding**. In: *NPGML Workshop*. NeurIPS. 2025.

[2] S. Ögüt, C. Vincent-Cuaz, MM, et al. **Benchmarking Instance-Level Learnability and Interpretability in Multiple Instance Learning**. In: Under Review. 2025.

[3] B. Chen*, C. Vincent-Cuaz*, L. A. Schoenpflug*, MM*, et al. **Revisiting Automatic Data Curation for Vision Foundation Models in Digital Pathology**. In: Early Acceptance (Top 9%). MICCAI. 2025.

[4] O. Zaghen, MM, L. Toni, and P. Frossard. **Graph Discrete Diffusion: a Spectral Study**. In: *DeLTa and XAI4Science Workshops*. ICLR. 2025.

[5] L. Sbicego, S. Ögüt, MM, Y. Qin, D. Thanou, and P. Frossard. **On the Role of Graph Structure in Graph Neural Networks**. In: *ICBINB Workshop*. Selected for PMLR Proceedings. ICLR. 2025.

[6] Y. Qin*, MM*, D. Thanou, and P. Frossard. **DeFoG: Discrete Flow Matching for Graph Generation**. In: Oral Presentation (Top 1%). ICML. 2025.

[7] MM, C. Vignac, D. Thanou, and P. Frossard. **Generative Modelling of Structurally Constrained Graphs**. In: NeurIPS. 2024.

[8] MM, D. Thanou, and P. Frossard. **Tertiary Lymphoid Structures Generation through Graph-based Diffusion**. In: *GRAIL Workshop*. MICCAI. 2023.

[9] S. Moalla*, MM*, L. Riccio*, and J. Lee*. **[Re] Reproducibility Study of Behavior Transformers**. In: *ML Reproducibility Challenge 2022*. Outstanding Paper Award (Honorable Mention). ReScience C. 2023.

[10] MM, R. Negrinho, J. Xavier, and P. M. Aguiar. **COCO Denoiser: Using Co-Coercivity for Variance Reduction in Stochastic Convex Optimization**. In: *OPT Workshop*. NeurIPS. 2021.

TALKS

ICML 2025, Vancouver, Canada, 2025. *DeFoG: Discrete Flow Matching for Graph Generation*.

AI Alliance: AI in Materials & Chemistry Webinar Series, [online](#). *DeFoG: Discrete Flow Matching for Graph Generation*.

Inductiva Machine Learning for Science & Engineering Summer School, Porto, Portugal, 2025. *An Introduction to Graph Neural Networks*.

BioML Workshop, Lausanne, Switzerland, 2025. *Constrained Graph Generative Models in Science*.

Applied Machine Learning Days, Lausanne, Switzerland, 2025. *Constrained Graph Generative Models in Science*.

DataMakers Fest, Porto, Portugal, 2023. *Generative Models in Science: A Peek into Graph Diffusion Models for Digital Pathology*.

TEACHING EXPERIENCE

Teaching Assistant for:

- Network Machine Learning (Lead TA in 2024/2025)
- Probability and Statistics
- Practice of Object-Oriented Programming
- Analysis I
- Histology

Supervised 5 students (4 MSc level, 1 junior PhD).

SOFTWARE SKILLS

LANGUAGES: Python, Matlab, Java, Mathematica, R

FRAMEWORKS: PyTorch, Keras, Tensorflow, Pandas, Scikit-learn, NumPy, CVX, CVXPY/CVXOPT, Abaqus

MISC: Unix, Git, Docker, Kubernetes, SLURM, Run:AI, AWS (EC2, S3)

EXTRACURRICULAR ACTIVITIES

2023 Attended Cambridge Ellis Unit Summer School on Probabilistic Machine Learning

2022 - 2024 Co-organizer of [Deep Learning Sessiong Portugal](#)

2021 Attended Lisbon Machine Learning Summer School

2009-21 Competitive football player (national level)

2015 National finalist in Biology Olympiad

LANGUAGES

PORTUGUESE: Native

ENGLISH: C2, TOEFL iBT: 114 / 120

SPANISH: A2

FRENCH: A2

CHINESE: Elementary (Tsinghua course)

PERSONAL INTERESTS

Football, CrossFit, Ski, Reading, Cinema

COMMUNITY SERVICE

Reviewer for NeurIPS, NPGML workshop (NeurIPS) and ComBayNS workshop (IJCNN)